National University of Computer and Emerging Sciences Islamabad Operating System Lab Spring 2025

Lab Task 1
Basic Linux Commands, File Permissions, Cond

Practice Resource

To help you practice shell scripting, refer to the following link with practical examples:
Shell Script Examples - GeeksforGeeks

This resource provides hands-on examples of shell scripts that will enhance your understanding of the topics covered in this lab. Ensure you explore and try out the examples before attempting the tasks**.**

Basic Linux Commands: Lab Tasks

Task 1: Create and Display Files

1. Create a new file named `activity_log.txt`.
2. Add the text "Exploring Linux Features" to the file.
3. Display the contents of the file in the terminal.

Task 2: Directory and File Management

1. Create a directory named `linux_tasks`.
2. Move the `activity_log.txt` file into the `linux_tasks` directory.
3. Navigate into the `linux_tasks` directory and list all files.

Task 3: File Deletion

1. Create three files named `doc1.txt`, `doc2.txt`, and `doc3.txt`.
2. Delete `doc1.txt`.
3. Verify that only `doc2.txt` and `doc3.txt` remain in the directory.

Linux File Permissions: Lab Tasks

Task 1: File Permissions Modification

1. Create a file named `notes.txt`.
2. Modify its permissions to:

    - Allow the owner to read and write only.
    - Allow the group to read and execute only.
3. Verify the changes using `ls -l`.

Task 2: Adding and Removing Permissions

1. Create a file named `config_data.txt`.
2. Grant read and execute permissions to everyone (user, group, and others).
3. Remove execute permissions for the user only.
4. Verify each change using `ls -l`.

Conditional Task: Elif Statement

Question:
Write a shell script to:

1. Ask the user to enter their age.
2. Check if the age falls into one of the following categories:
    - Less than 13: Print "Child".
    - Between 13 and 19 (inclusive): Print "Teenager".
    - Greater than 19: Print "Adult".
3. Display an appropriate message based on the age using `elif`.

Loop Task: For Loop

Question:
Write a shell script to:

1. Prompt the user to enter a number.
2. Use a `for` loop to print the square of each number from 1 to the entered number.
3. Display "All squares calculated!" once the loop completes.

Instructions for Students

1. Complete all tasks in sequence.
2. Write the required scripts for conditional and loop tasks in separate files.
3. Test each task in the terminal and document the results.
4. Submit your completed scripts and terminal outputs for review.

THE END